

Assignment: NET Centric Computing

Course Code: CSC378

Topic: Unit 3 - HTTP and ASP.NET Core (Theory & Practical)

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Due Date: 24 August

Submission Instructions:

Please complete the following questions by combining theoretical explanations with practical implementations. For the practical components, include the required commands, code snippets, project structure details, and screenshots of your application execution and testing results.

Questions:

1. HTTP Communication and ASP.NET Core Web API Development

Explain HTTP and describe the HTTP request and response message formats with suitable examples. Discuss different HTTP methods used in web applications and explain the role of client-server architecture in modern web applications.

Develop an ASP.NET Core Web API application to demonstrate HTTP communication between client and server. Perform the following tasks:

1. Create a new ASP.NET Core Web API project using .NET CLI.
2. Design a model class (such as Student, Employee, or Product) with appropriate attributes.
3. Implement RESTful API endpoints using HTTP methods:
 - GET – Retrieve data from the server.
 - POST – Add new data to the server.
 - PUT – Update existing data.
 - DELETE – Remove data from the server.
4. Test the implemented APIs using Swagger or Postman.
5. Analyze and document:
 - HTTP request message format.
 - HTTP response message format.
 - HTTP status codes.
 - Client-server request and response flow.

6. Include screenshots showing:

- API endpoints.
- HTTP request and response messages.
- Successful API execution results.

2. ASP.NET Core Architecture, MVC Pattern, Projects, and Application Development

Explain the ASP.NET Core architecture and its major components including Kestrel Server, Middleware Pipeline, Routing, Dependency Injection, and Configuration System. Describe the MVC pattern and explain how ASP.NET Core MVC helps in developing dynamic, maintainable, and scalable web applications. Also explain ASP.NET Core project structure and conventions.

Develop an ASP.NET Core MVC application to demonstrate MVC architecture and ASP.NET Core application structure. Perform the following tasks:

1. Create an ASP.NET Core MVC application using .NET CLI.

2. Design the application using MVC components:

- Create Model classes to manage application data.
- Create Controllers to handle user requests.
- Create Views using Razor pages for user interaction.

3. Implement basic CRUD operations:

- Add records.
- Display records.
- Update records.
- Delete records.

4. Explore and explain important ASP.NET Core project files and folders:

- Program.cs
- Controllers folder
- Models folder
- Views folder
- wwwroot folder
- appsettings.json

5. Implement custom middleware to demonstrate HTTP request processing.

6. Configure Dependency Injection by:

- Creating a custom service.
- Registering the service.
- Using the service inside a controller.

7. Run the application and demonstrate the ASP.NET Core request processing pipeline.

8. Document:

- ASP.NET Core architecture flow.
- MVC request-response flow.
- Project structure and conventions.
- Middleware execution.
- Dependency Injection implementation.

9. Include screenshots showing:

- MVC project structure.
- Running application.
- CRUD operations.
- Middleware output.
- Service configuration.